

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from:</i> • Raise river banks / build sea walls, flood barriers or wave screens • Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff • River flow management: upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff • Raised buildings on stilts or floating houses • Pumping rainwater from land below river level • Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh)	3
01.2	• Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective • Land is abandoned as water levels rise / coast erodes; defences used only where land is very important	2
01.3	<i>Any 2 from:</i> • Road surfaces deform as tar melts - need tarmac with a higher melting point • Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress • Bridge damage from high river flows / debris; landslides on deforested slopes	2
01.4	• CO₂ is captured from the flue gases (or carbon removed from the fuel before combustion) • Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery • Limitation: developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
02.1	The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically)	1
02.2	<i>Any 4 from:</i> - Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions - Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition - Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils - pH / presence of other chemicals may increase or decrease the rate of degradation - Adsorption onto clay / organic matter may protect from degradation and increase persistence	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
03.1	<i>Any 3 from:</i> - Stress-related illness: ulcers, high blood pressure, heart disease - Behavioural changes: irritability, aggression, lack of concentration leading to accidents - Communication problems / educational underperformance (eg schools under flight paths) - Sleep disturbance	3
03.2	- Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) - Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes - Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry	3
03.3	- Birds that sing to establish territories / attract mates cannot be heard against background noise - Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea	2
03.4	- Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source - Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	
04.1	- CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again - During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls - In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises - The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large	4

Question	Marking guidance	Marks
04.2	- Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) - So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean	2
04.3	<i>Any 2 from:</i> - Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) - Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) - Atmospheric pressure allows liquid water to exist - Distributes heat from the tropics to higher latitudes by winds - Transports water vapour to areas that would otherwise be dry	2
04.4	- Troposphere - Stratosphere	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	
05.1	- The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) - High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) - High energy density fuels reach high temperatures needed for smelting / steam turbines	3
05.2	$500\,000 / 28 = 17\,857\text{ kg}$ $= 17.9\text{ tonnes}$ (accept 18; book quotes 13 t for 0.7% enriched fuel)	2
05.3	- The electricity they generate can power high-energy-requirement technologies - Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage)	2
05.4	- Intermittent: not available all the time, eg tidal (but the times are predictable so alternatives can be planned) - Unpredictable: intermittent and the timing / amount cannot be forecast accurately, eg wind, solar	2
05.5	Gaseous natural gas has a very low energy density by volume; LNG at -160 °C has 160 times the energy density so a ship can carry a worthwhile amount	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	
06.1	- Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve - Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed - Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle	3
06.2	- Auger collects a core of known cross-sectional area and depth - So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume)	2

Question	Marking guidance	Marks
06.3	- Minerals such as calcium carbonate break down releasing carbon dioxide - So mass is lost that is not organic matter	2
06.4	- A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter - Standardise bulb brightness / colour, distance and container - A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
07.1	<i>Any 3 from:</i> - Volume of ash is much less than the original waste (less landfill) - Heat used for district heating / electricity generation (energy recovery) - No sorting / complicated management needed - Destroys pathogens / flammable organic wastes converted to CO₂ and water - Refuse-derived fuel can replace fossil fuels (eg cement kilns)	3
07.2	- High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) - Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH - Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value - Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost	4
07.3	- Ash may contain concentrated heavy metals / toxic residues - Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble	2
07.4	<i>Any 1 from:</i> - Fuel (natural gas) needed to burn wet / non-flammable wastes - Cost of treating effluent gases (SO₂, NO_x, HCl, smoke)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
08.1	- Mutagen: an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos - Carcinogen: a mutagen that causes cancer / uncontrolled cell division producing a tumour - Teratogen: an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T	3
08.2	- Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on - Mutagens change DNA; if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited	2

Question	Marking guidance	Marks
08.3	• Somatic: mutation in a general body cell, affecting the individual only (eg cancer / tumour) • Gonadic: mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring	2
08.4	• Two or more pollutants whose effects interact to produce a different (usually more serious) effect • The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) • Example: ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells; smoke and sulfur dioxide in smogs; neonicotinoids with fungicides	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	
09.1	• As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically	1
09.2	• $2400 / 40 = 60$ times	1
09.3	• Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) • Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits • The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall	3
09.4	• Lowering the cut-off ore grade includes deposits of lower purity in the reserves • Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%)	2
09.5	• The total quantity (stock / resource base) is far greater than could ever be used • But most is at purities too low or in locations too inaccessible for current economic exploitation • The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable; energy needs rise rapidly as ore grade falls	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable; demand fluctuates daily, weekly, seasonally and short-term; supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants; large-scale electricity storage not yet possible directly so energy must be converted • Pumped-storage HEP: proven, rapid response, peak shaving, but limited sites • Batteries: lithium-ion for grid and vehicles; factors affecting viability; V2G • Hydrogen: electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid; high energy density enables vehicle use • Compressed air with heat recovery; thermal storage (molten salt for CSP, high-volume stores, thermal mass); flywheels; supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage; energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage; judgement	9
10.2	• Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25